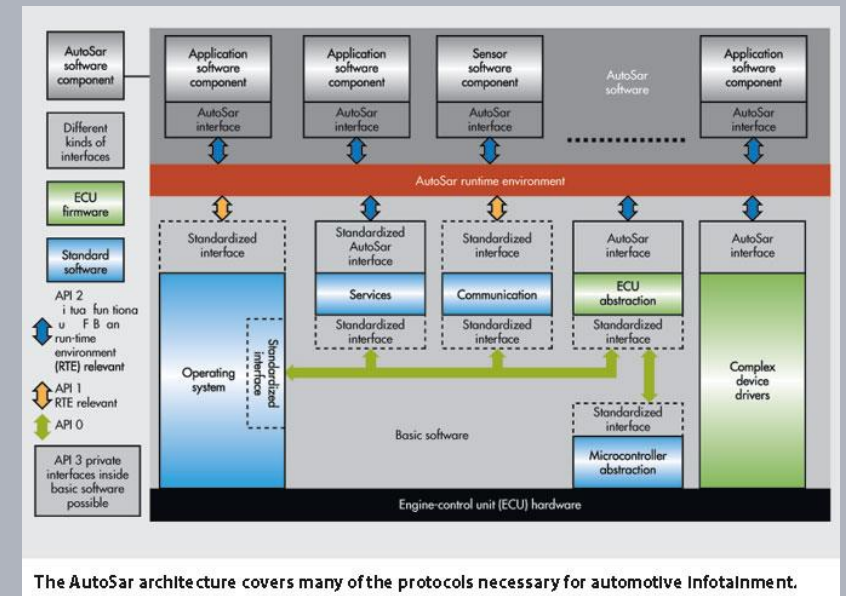


Advanced Microcontroller Architectures and Embedded OS

Introduction

Prof. Dr.-Ing. Peter Fromm

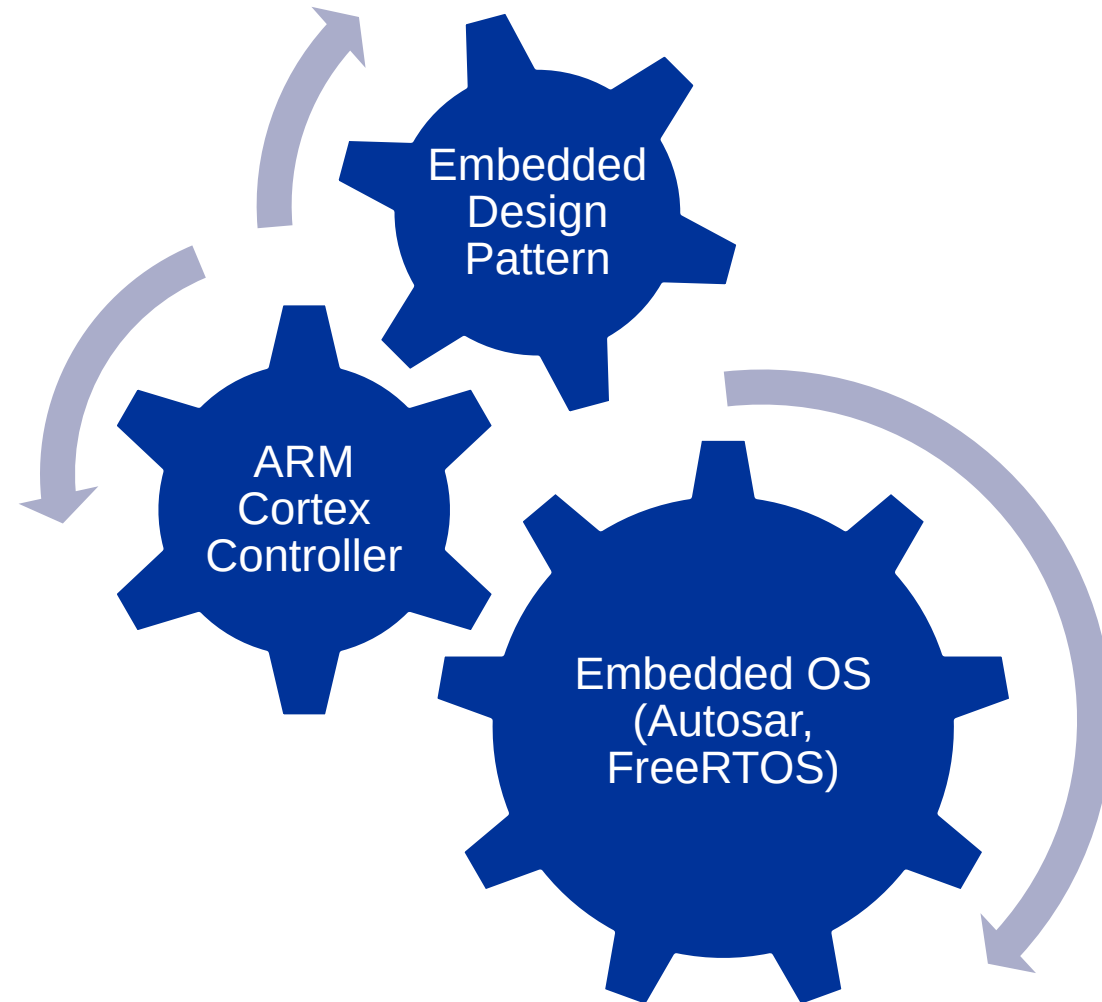


The AutoSar architecture covers many of the protocols necessary for automotive Infotainment.

Content

- Course Goal
- Content
- Lab Organisation

This course in a Nutshell



Goal of this module

Content of the course Advanced Micro-Controller Systems and Embedded Operating Systems

- Hardware Architecture of current Micro-Controller Systems
- RTOS implementation
- Tasks, Events, Messages, Semaphores
- Critical Sections , Priority Ceiling, Deadlocks
- Scheduling algorithms
- Safety and Memory-Protection
- Hardware Security Features
- Hardware Device-Driver development
- Efficient Implementation of DSP algorithms

Content of the lab

- Design and implementation of deep embedded software on a 32-bit Micro-controller
- Configuration and application of embedded operating system services

Learning outcome

to understand:

- Distinct features of state of the art Micro-Controllers

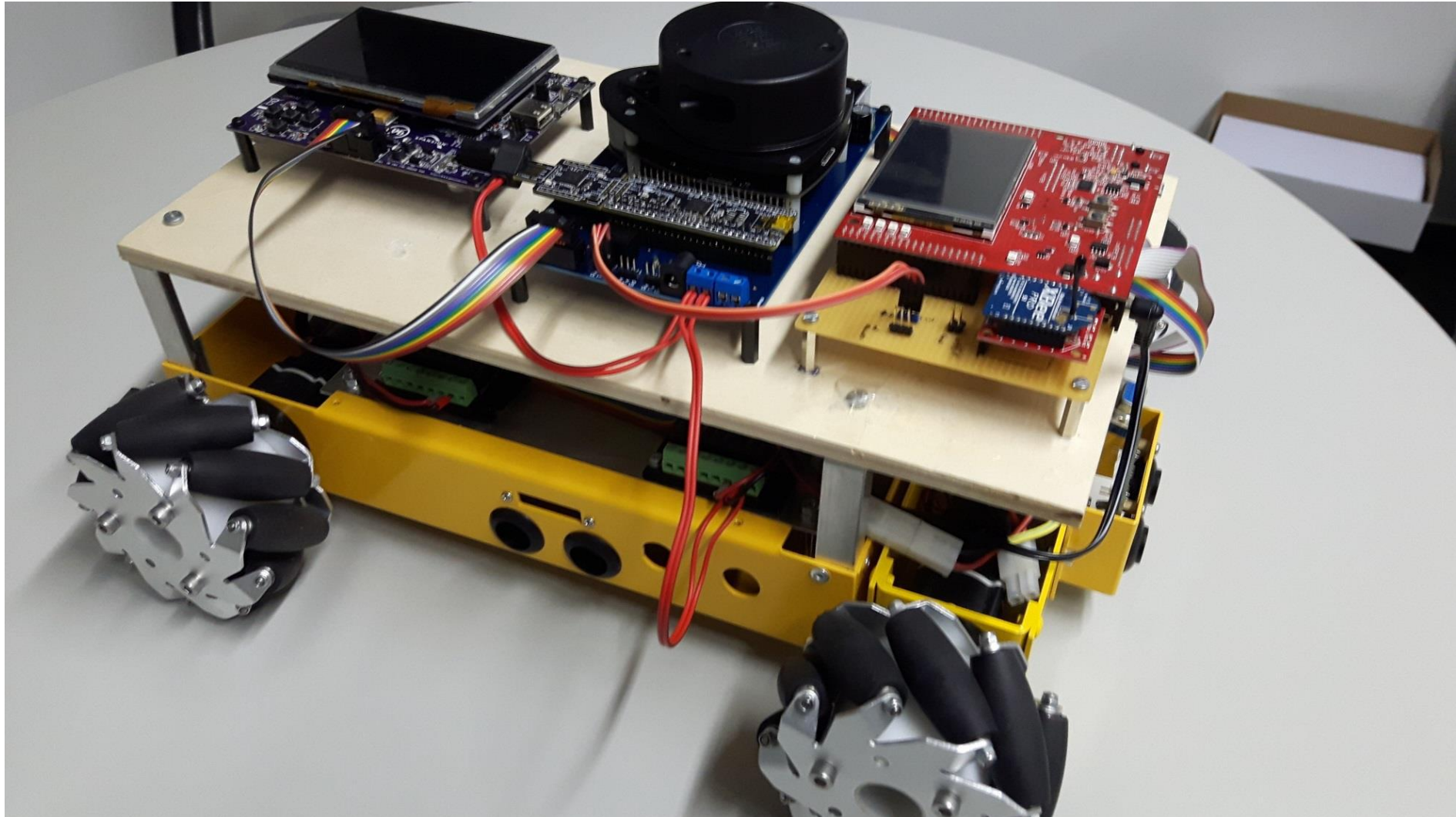
to apply:

- Design and Implementation of Device-Drivers
- DSP-Algorithms and Control Software for deeply Embedded Applications.

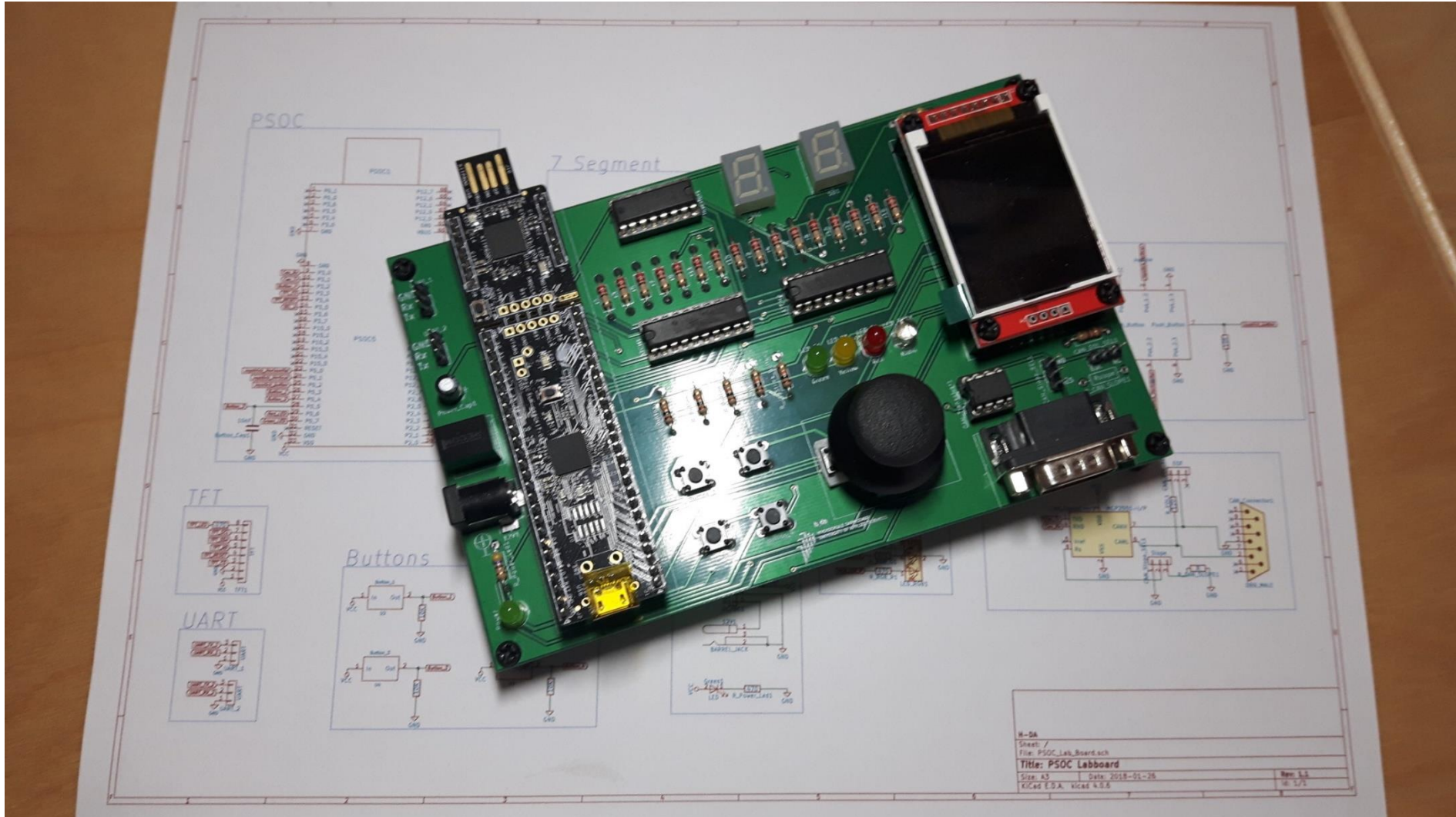
Structure of the Lecture

Week	Lecture I (Schaefer 2SWS)	Lecture II (Fromm 1SWS)
1	Microcontroller Basics	Embedded OS, Multitasking
2	Cortex Architecture I	
3	Cortex Architecture II	PSOC Example Application
4	Interrupts	
5	OS Basics / MyOS (Scheduler)	OSEK/Autosar OS API
6	Scheduling / Multithreading	
7	FreeRTOS	Inter Task Communication
8	Memory Protection	
9	Watchdog, Alive Monitoring	State Machine Basics
10	Algorithms / Efficiency I / State Machine	
11	Algorithms / Efficiency II	Critical Sections, Priority Ceiling Protocol, Deadlocks
12	Performance	
13	Exam Preparation	

Toys for big boys and girls



LabBoard



Lab Organisation – OSEK / AUTOSAR

Mandatory assignments

- Reaction Game with Autosar-OS/Erika
- Smart Volume

Lab execution

- Design and implementation as lab@home / Anydesk
- Labs will be reviewed and must be passed (exam prerequisite)
- Lab dates (reviews) will be communicated

Lab content will be part of the exam!

